Investigation report

Did newer signup cohorts convert to subscription at a lower rate?

Compare mature signup cohorts on the approved 30-day subscription conversion metric.

Data

Authorized PocketToons SQLite synthetic fixture
April-August 2026 signup cohorts, UTC

Snapshot

users_snapshot_at 2026-09-30T23:59:59Z (asserted dataset metadata)
Build 3a0dbed21d2f45958aefa2887a0cadd8

Run

Independent
2026-11-09
7 logged checks

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How to read this report

- Checks refer to separate JSONL evidence logs with SQL and results.
- Gold is a precomputed cohort summary; Silver supplies user-level cohort details.
- Observed associations do not establish causes.

Terms used in this report

Term	Meaning
Signup cohort	All users who signed up in a UTC calendar month.
30-day conversion	Eligible users with a successful subscription charge in the 720 hours starting at signup, divided by eligible users.

01. Chapter

The answer

01

Yes. Every mature cohort after April converted below April, and July-August cohorts were particularly low. The pattern is not strictly monotonic; September is not yet measurable on a complete 30-day basis.

2.59%	0.53%	-2.05 pp
April-June 167 / 6,454	July-August 29 / 5,426	Difference 79% relative decline

- April was 4.56%, May 1.73%, June 2.11%, July 0.52%, and August 0.54% (check 3).
- Rates fell within organic, paid search, and paid social; the acquisition mix also shifted strongly toward paid social (check 5). This does not identify a cause.
- Investigate checkout and subscription payment pathways and validate later cohorts when fully observed.

02. Chapter

How we investigated

02

What data we used

- Approved funnel and lifecycle group v1.0; charge success time in [signup, signup + 30 days), including charges subsequently refunded. Cohort includes all eligible signups, active or inactive.
- Coverage is [2026-04-01, 2026-10-01) UTC per repository contract; windows must end on or before coverage end. March signups and 1,720 September signups are excluded from this metric (checks 3 and 7).
- Gold cohort month-zero rows avoid repeated cohort-level counts. Silver profile supports acquisition slices and matches Gold totals (checks 3 and 5).

03. Chapter

What happened

03

The movement

Signup month	Eligible	Converted	Rate
April 2026	1,600	73	4.56%
May 2026	2,201	38	1.73%
June 2026	2,653	56	2.11%
July 2026	2,477	13	0.52%
August 2026	2,949	16	0.54%
September 2026	0 of 1,720	Not observed	Not available

Who drove it

Acquisition source	April-June converted / eligible	July-August converted / eligible
Organic	82 / 3,204 (2.56%)	6 / 996 (0.60%)
Paid search	38 / 1,697 (2.24%)	5 / 907 (0.55%)
Paid social	44 / 1,489 (2.96%)	18 / 3,469 (0.52%)
Unknown	3 / 64 (4.69%)	0 / 54 (0.00%)
All	167 / 6,454 (2.59%)	29 / 5,426 (0.53%)

- Pooled rates use summed users and converters, not an unweighted average of monthly rates. Paid social increased from 23.1% to 63.9% of eligible signups (check 5).
- The decline within each major source shows the overall decline is not solely explained by changing source weights. These are descriptive comparisons, not causal effects.

04. Chapter

Why it happened

04

Hypothesis	Status	Evidence
Lower conversion across acquisition sources	Supported pattern	All three major sources fell between pooled periods (check 5); reason unknown.
Lower seven-day activation explains decline	Not supported as overall pattern	Activation was 53.0% versus 54.4%; this does not rule out downstream or subgroup issues (check 5).
Acquisition mix alone explains decline	Rejected as sole arithmetic explanation	Each major source also declined; mix changes may still contribute (check 5).

05. Chapter

Data quality and limits

05

Data quality

- Build status succeeded, with build-time Gold cohort and Silver profile conversion reconciliation recorded as passing (check 2). Snapshot metadata is an asserted timestamp, not a verified database hash or freshness guarantee.
- All 24,000 profiles have a non-null user and signup; no duplicate user IDs. There are 13 reviewed payment corrections across the dataset (check 4). Two corrected subscription transactions fall in eligible conversion windows, one each in July and August (check 6). Removing at most those two converters would make later rates even lower; this is a conservative sensitivity, not a claim that corrections are wrong.
- The build records registry SHA-256 57b21d...1227; current registry file SHA-256 is 4b5b2f...5554. Both state lifecycle v1.0, but byte-level mismatch means exact build-time source text was not verified. Underlying source fingerprints were recorded by the build but not independently rehashed now.

Limitations

- The period split is a descriptive comparison chosen after observing monthly results; no randomized or causal conclusion follows.
- September is immature, and March lies outside the defined observation window. Observed 30-day success is not lifetime conversion or subscription access.
- Small converter counts, especially in individual slices, make rates unstable. Source attribution is first touch and unknown remains explicit.
- The repository is a downloaded copy without Git metadata. AGENTS.md filesystem creation and modification timestamp is 2026-11-09 20:12:48 IST; this is a proxy for download date, not a commit identifier.

06. Chapter

What to do next

06

01 Validate payment and checkout instrumentation and compare acquisition-source-specific funnel journeys for the affected cohorts; establish complete follow-up before comparing.

02 Track mature 30-day cohorts and relevant quality exceptions; investigate the build registry hash mismatch and verify source freshness before operational decisions.

03 Test candidate interventions experimentally before attributing the change to acquisition, onboarding, pricing, or payment friction.

Appendix

Evidence appendix

A

Check	Purpose	Evidence file
1	Layer inventory and schema	01_inventory.jsonl
2	Snapshot, build, definitions and reconciliation	02_metadata.jsonl
3	Monthly cohort conversion	03_cohorts.jsonl
4	Exceptions, nulls, duplicates and build IDs	04_quality.jsonl
5	Cohort source and activation breakdown	05_segments.jsonl
6	Reviewed payment correction sensitivity	06_corrected.jsonl
7	Profile coverage and incomplete windows	07_coverage.jsonl